



# Data-Driven Identification and Analysis of Waiting Times in Business Processes

## A Systematic Literature Review

Muhammad Awais Ali · Fredrik Milani · Marlon Dumas

Received: 3 January 2023 / Accepted: 12 March 2024 / Published online: 15 May 2024  
© The Author(s) 2024

**Abstract** Reducing waiting times in end-to-end business processes is a recurrent concern in the field of business process management. The uptake of data-driven approaches in this field in the past two decades, most notably process mining, has created new opportunities for fine-grained analysis of waiting times based on execution data. As a result, a wide range of approaches for waiting time identification and analysis on the basis of business process execution data have been reported in the literature. In many instances, different approaches have considered different notions of waiting time and different causes for waiting time. At present, there is a lack of a consolidated overview of these manifold approaches, and how they relate to or complement each other. The article presents a literature review that starts with the question of what approaches for identification and analysis of waiting time are available in the literature, and then refines this question by adding questions which shed light onto different causes and notions of waiting time. The survey leads to a multidimensional taxonomy of data-driven waiting time analysis techniques, in terms of purpose, causes, and measures. The survey identifies gaps in the field, chiefly a scarcity of integrated multi-causal approaches to analyze waiting times in business processes, and a lack of empirically validated approaches in the field.

**Keywords** Business process management · Process mining · Waiting time

## 1 Introduction

Business analysts continuously strive to improve business processes with respect to different performance measures. Typical performance measures relate to cost, time, quality, and flexibility (van der Aalst 2016; van der Aalst et al. 2016). A recurrent performance measure in the field of Business Process Management (BPM) is *cycle time* (Dumas et al. 2018): the time from the start of the first activity instance until the end of the last activity instance of a case.

The cycle time of a case is the sum of its processing time and waiting time. Processing time is the time resources spend performing activities in a case. Waiting time is the time during which activities of a case are not being processed. It is common that the waiting time in a business process is larger than the processing time (Dumas et al. 2018). In these situations, the cycle time of a process can be improved by addressing the various causes of its waiting times (Lashkevich et al. 2023). Given the prominent role of waiting times in business processes, business analysts are interested in identifying and quantifying waiting times, and analyzing their sources (Mannhardt et al. 2019; Nogayama and Takahashi 2015).

The causes (or sources) for waiting times in business processes are diverse. For example, waiting times may be caused by resource contention—a resource does not start an activity instance as it is busy with another activity (Lashkevich et al. 2023). Another cause of waiting time can be resource unavailability—a resource is off-duty (Toosinezhad et al. 2020). Similarly, in healthcare

---

Accepted after three revisions by Manuel Resinas.

---

M. A. Ali (✉) · F. Milani · M. Dumas  
University of Tartu, Tartu, Estonia  
e-mail: muhammad.awais.ali@ut.ee

F. Milani  
e-mail: fredrik.milani@ut.ee

M. Dumas  
e-mail: marlon.dumas@ut.ee

processes, an internal cause of waiting time may be a patient waiting for the next doctor to become available (Aissaoui et al. 2022). Another example of a waiting time is when in a process we wait for a response from a customer or a produced item needs to cool down for a period of time before proceeding to the next step in a process (Satitcharoenmuang et al. 2017).

Despite extensive research in process mining, a unified perspective on different notions of waiting times has yet to be established. For example, some authors define waiting time as the time during the execution of a case when no activity is being performed, also known as “idle time” (Drosouli et al. 2020). In contrast, other authors define waiting time as any time between the end of the current activity and the start of the next activity regardless of other activities executed during this waiting time (Senderovich et al. 2019; Leemans et al. 2018). Similarly, several studies direct their focus to waiting times that originate from resource contention (Lashkevich et al. 2023) but neglect other factors such as batching (Lashkevich et al. 2022), resource unavailability (Aissaoui et al. 2022), or prioritization (Rojas et al. 2018; Cho et al. 2019). Some studies also highlight waiting times arising from external sources, such as waiting for a response from a customer (Ferreira and Vasilyev 2015) or patients waiting for the treatment to take effect (Jaisook and Premchaiswadi 2015; Mans et al. 2012), contributing to waiting times in business processes.

These studies have considered different notions and causes of waiting times by relying on different approaches for quantifying waiting times. Therefore, there is a need for a taxonomy that encapsulates how waiting times manifest themselves in business processes, addressing both their causes and implications. In this paper, we conduct a systematic literature review to propose a multi-dimensional taxonomy of different types and sources of waiting times in a business process. More specifically, the proposed taxonomy offers conceptual clarity by detailing the different types of waiting times, their sources, and how they can be analyzed using data-driven methods. Furthermore, our proposed taxonomy provides terminological clarity by categorizing various approaches employed for quantifying waiting times in business processes. Additionally, it provides a unified perspective on how the proposed approaches for waiting times analyses can be validated. While our systematic literature review shares some common ground with existing reviews regarding delay modeling and predictive process monitoring, it distinguishes itself by specifically addressing the various types and sources of waiting times within business processes.

The remainder of this paper is structured as follows: Sect. 2 presents background and related research concerning the identification and analysis of waiting times. In Sect. 3, we outline our approach for conducting a

systematic literature review. Section 4 presents the findings, and Sect. 5 discusses the implications of the findings. Finally, Sect. 6 draws conclusions and outlines possible avenues for future work.

## 2 Background and Related Work

This section presents background and positions our work in the context of existing research. More specifically, we introduce business processes and performance measures for business processes: process mining, and discuss other systematic literature review studies within the field of process mining.

### 2.1 Business Processes and Process Performance Measures

A business process can be defined as a set of activities that take some input and produce some output to add value to an organization and its customers (Dumas et al. 2018). Activities in business processes are discrete units of work that detail the logical steps within a process. Every execution of a business process results in a process instance, which relates to a specific case. For example, in an ‘order-to-cash’ process, each order triggers a unique process instance, and that order itself can be referred to as a *case*. During the execution of this process instance (or case), multiple activities are performed. Each execution of these activities within the process is called an *activity instance* that is associated with that particular case. Each activity instance is executed by an entity which could be a human participant, a machine, or even a software service, all commonly referred to as *resources*.

The performance of a business process can be measured with consideration to the dimensions of cost, time, quality and flexibility (Dumas et al. 2018; van der Aalst 2016). These performance dimensions are part of the so called Devil’s Quadrangle (Reijers and Mansar 2005; Mansar and Reijers 2007). The Devil’s Quadrangle suggests that improving one dimension of a process can potentially negatively impact the performance of another dimension. For instance, if the time dimensions—i.e., the efficiency or speed of a process—have improved due to increased standardization, it will be at the expense of flexibility. As such, the performance dimensions are inter-dependent. Furthermore, Milani and Maggi (2018) propose a framework for categorizing process mining techniques which measure performance of business processes. This paper suggests that the dimensions of time can be measured with a variety of metrics, such as process duration, fragment duration, activity duration, waiting duration, and delay duration. A commonly referenced temporal measure of

process performance is *cycle time*, i.e., the time between the beginning and end of a case in a process (van der Aalst et al. 2016). For instance, the cycle time of a loan application process spans from the moment a customer submits the loan application until the application is either accepted or rejected.

Cycle time is usually determined by its two constituent components, namely *Processing Time* and *Waiting Time*. *Processing Time* is the time that a resource spends on executing an activity instance. The remaining time of the cycle time is called *Waiting Time*.

## 2.2 Process Mining

Process mining is a family of data-driven techniques that can aid analysts to gain insights regarding business process behavior from different perspectives Milani et al. (2022), such as discovering process models (van der Aalst 2011), assessing conformance of recorded process behavior to a baseline model (Dunzer et al. 2019), analyzing performance (Diba et al. 2019), conducting predictive (Francescomarino et al. 2018) and prescriptive (Kubrak et al. 2022b) monitoring, and implementing process improvements (van der Aalst 2016; Kubrak et al. 2022a).

The input for process mining techniques is an event log extracted from an enterprise information system that records information about every case of a process. An event log consists of a set of events, each of which captures an activity instance of a case, resource, timestamps information, and case id. To illustrate this, consider a loan origination process. An event might be ‘loan application received,’ which captures the activity of receiving a loan application, along with the resource that recorded it and its timestamp. Each event can be grouped according to its uniquely assigned case identifier. In the loan origination process example, each of these events belongs to a unique loan application, identified by, for instance, its loan application ID. A single activity instance can be captured by multiple events. For instance, a “start event” and a “completion event” might both relate to one particular activity instance. In our loan origination example, the process might start with the ‘loan application received’ event and end with either a ‘loan approved’ or ‘loan rejected’ event. Moreover, the sequence of events in a case forms a trace, and a set of complete traces (i.e., traces indicating the completion of cases) forms an event log. To continue with our example, as the loan application progresses, it generates a series of events, creating an execution trace. Multiple such traces from various loan applications collectively form an event log. While the process execution data underlying these event logs originate from an information system, it should be noted that

event logs are constructed rather than directly extracted from the information system.

Process analysis tools (e.g., ProM, Disco, Apromore, and Celonis) use process mining to aid process analysts to quantify and assess the performance of a business process by overlaying the process model (often referred to as BPMN models or process maps) with performance diagnostics such as cycle time, processing time, and waiting time (Capitán-Agudo et al. 2022). With this enhanced process analysis, a process owner can analyze the cycle time of a process by identifying waiting time bottlenecks and implementing strategies to improve business processes by reducing or mitigating delays.

## 2.3 Related Work

Existing surveys in process mining focus on the specific capabilities of process mining techniques. The authors of Dumas et al. (2018) classified process mining tools into four categories: Automated Process Discovery, Conformance Checking, Variant Analysis, and Performance Analysis. Several review studies have examined each of these categories. For instance, in Augusto et al. (2019), the authors conduct a systematic literature review on discovering process models from event logs and empirically compare these techniques against benchmarks. Similarly, a survey of conformance checking (Dunzer et al. 2019) discusses techniques for comparing the behavior observed in an event log against the behavior captured in a process model. Another review study (Taymouri et al. 2021) examines process mining for variant analysis, i.e., comparing two or more variants of a business process using event logs. Finally, in Milani and Maggi (2018), the authors propose a framework for categorizing temporal performance analysis using process mining. Thus, existing literature review studies either do not cover the temporal perspectives of business processes. When they do, as in Milani and Maggi (2018), the analysis is limited to cataloging different types of temporal performance measures. In contrast, our paper focuses on cycle time as a temporal performance metric and its analysis using process mining techniques.

Process mining has expanded to encompass predictive and prescriptive capabilities. In this regard, several surveys have been conducted on various aspects of predictive process monitoring, including outcome prediction (Teinmaa et al. 2019), time prediction (Verenich et al. 2019) and activity sequence prediction (Rama-Maneiro et al. 2023) of a case in a business process. The analysis of cycle time has also been approached in studies on predictive business process monitoring (Lamghari et al. 2019; Francescomarino et al. 2018; Verenich et al. 2019), proposing methods to predict the remaining time of an

ongoing case in a business process. There has also been a survey on prescriptive process monitoring (Kubrak et al. 2022b) that proposes a framework for characterizing prescriptive process monitoring methods according to their performance objective, performance metrics, intervention types, modeling techniques, data inputs, and intervention policies. Although existing works include cycle time, they do not analyze it in more detail within the context of process mining techniques.

### 3 Review Method

The objective of this paper is to examine quantitative and qualitative data-driven methods and techniques, with a particular focus on process mining, for identifying and analyzing waiting times in business processes. To achieve this objective, we employ a Systematic Literature Review (SLR) as it allows for collecting relevant studies, and based on their examination, we identify and classify approaches to analyze waiting times in business processes. We followed the guidelines proposed by Keele et al. (2007). According to these guidelines, we developed a review procedure consisting of three main phases: Planning, Conducting, and Reporting. This section explains the steps we followed in the first two phases, while the next section corresponds to the third phase.

#### 3.1 Planning the Review

We initiated our study by formulating research questions and developing the review protocol based on Keele et al. (2007). We address the following initial research question:

*RQ1* What objectives do existing data-driven approaches aim to achieve, and what techniques do they employ to identify and analyze waiting times in business processes?

In general, the goal of an analyst, when identifying and analyzing waiting times, is to find ways to improve business processes (Kubrak et al. 2022a). To this end, the analyst needs to identify the cause or source of waiting times. Accordingly, this literature review also addresses the following research question:

*RQ2* What sources of waiting times are these approaches designed to analyze?

Furthermore, analysts aim at quantifying waiting times by specifically identifying the various ways the notion of waiting times, such as idle time and simple waiting time, manifest themselves in a business process (Kubrak et al. 2023). In line with this perspective, our literature review also addresses the following research question:

*RQ3* How do the identified approaches measure waiting times in business processes?

Different approaches for identifying and analyzing waiting times have been validated to different extents using different methods. Arguably, the validation of the proposed methods is essential in asserting their fit-for-purpose and potential usefulness. Accordingly, this literature review additionally addresses the following research question:

*RQ4:* How have the identified approaches been validated?

To address the aforementioned research questions, the review protocol consists of developing search strings, identifying relevant databases, defining inclusion and exclusion criteria, and specifying a data extraction strategy. Access to the complete review protocol is provided as supplementary material.<sup>1</sup> We followed the guidelines proposed by Keele et al. (2007) for constructing the search string. We derived keywords from the research questions to define the search strings. We divided the search string into two parts, namely string S1 and string S2. String S1 limits the investigation to methods that rely on event logs. Similarly, string S2 includes keywords related to waiting time and its synonyms found in the literature, such as delay (Mannhardt et al. 2019), shelf-time (Andrews and Wynn 2017) and idle (Drosouli et al. 2020). Both strings S1 and S2 were combined using logical operators to create a search string. The resulting search string we used is as follows. (“event log ”OR “process mining ”OR “workflow mining ”) AND (“waiting time ”OR “delay ”OR “shelf time ”OR “idle ”)

We applied the search string to ACM Digital Library, Scopus, Web of Science, and IEEE Explore to identify potentially relevant papers (see Table 1). We selected these databases because they collectively index most publications related to process mining.

According to the guidelines, we developed a set of exclusion and inclusion criteria to assess the relevance of studies (see Table 2). We began by excluding papers that were inaccessible (EC1) and non-English publications (EC2). Duplicates, i.e., papers with the same title and author, were also filtered out (EC3). Furthermore, we rejected papers that were published in non-peer-reviewed venues (EC4).

Following the application of the exclusion criteria, the remaining studies were evaluated using inclusion criteria. More specifically, the first inclusion criterion (IC1) covers research that addresses or illustrates a theoretical or practical approach, method, or technique to identify or analyze waiting times in business processes. The second inclusion

<sup>1</sup> <https://doi.org/10.5281/zenodo.10698027>.

**Table 1** Search Sources

|                   |                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Digital Libraries |                                                                                                                                    |
|                   | ACM Digital library                                                                                                                |
|                   | IEEE Xplore                                                                                                                        |
|                   | Web of Science                                                                                                                     |
|                   | Scopus                                                                                                                             |
| Searched Items    | Journal, workshop and conference papers                                                                                            |
| Search applied on | Full text—to avoid missing articles that do not have our search terms in their titles or abstracts but are relevant to the review. |
| Language          | English                                                                                                                            |

**Table 2** Exclusion and Inclusion Criteria Applied

|                    |                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Exclusion Criteria |                                                                                                                              |
|                    | EC1: The paper is digitally inaccessible                                                                                     |
|                    | EC2: The paper is written in language other than English                                                                     |
|                    | EC3: <i>The paper is a duplicate</i>                                                                                         |
|                    | EC4: The paper is published in a non-peer-reviewed journal or other source                                                   |
| Inclusion Criteria | IC1: The paper presents/demonstrates a method, approach or technique to identify/analyze waiting times in business processes |
|                    | IC2: The method, approach or technique presented/demonstrated in the paper uses event logs                                   |

criterion (IC2) requires that the paper’s presented method, approach, or technique specifically utilize event logs. In this context, an “approach” encompasses a set of ideas proposed for problem-solving. A “method” denotes a standardized procedure employed consistently to achieve a specific goal, which may integrate multiple techniques. A “technique” refers to a distinct procedural step within a method, tailored to address a specific facet of a problem. Furthermore, to ensure comprehensiveness and reduce the risk of overlooking relevant papers, we employed backward referencing Okoli (2015). Specifically, after identifying papers that met all the inclusion criteria, we scrutinized their reference lists and applied the same inclusion and exclusion criteria to identify relevant articles. The details of studies obtained via backward referencing are documented in “primary\_secondary\_search\_results.xlsx,” found as supplementary materials.

Finally, we designed a data extraction form (see Table 3). According to Okoli (2015), Brereton et al. (2007), a data extraction form enables systematic, unbiased, and consistent data collection. The data extraction form was designed by first collecting information about each publication (title, authors, year of publication, and

number of citations). Then, we specified the data needed to answer our research questions. For (RQ1), we extracted data such as “data driven techniques” and “objectives” whereas for (RQ2), we extracted data related to the source of waiting times. Then, addressing (RQ3), we extracted data needed for describing different notions for quantifying waiting times in business processes. Finally, for (RQ4), we included information on how existing approaches for identifying and analyzing waiting times in business processes are validated.

### 3.2 Conducting the Review

We conducted the search on 21 August 2022 and our search yielded 2611 hits, as shown in Table 4. We filtered the initial list of papers by using EC1 (inaccessible), EC2 (language), and EC4 (non-peer-reviewed). This resulted in the elimination of 173 studies. The remaining 2438 publications were further screened based on the EC3 (duplicate) resulted in the elimination of additional 253 studies. The remaining 2185 studies were screened based on title and abstract (IC1), resulting in 235 papers remaining as potentially relevant for our study. Finally, we applied IC2 by reading the entire text. This resulted in the filtering out additional 130 studies. As a result, 105 papers remained. Furthermore, 7 publications were added through backward referencing, yielding a final list of 112 relevant research articles. We used PRISMA<sup>2</sup> diagram (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) for the filtering process. This diagram is included as supplementary material.

## 4 Results

In the subsequent subsections, we report the results of our review. We begin with a quantitative overview of the identified studies. Next, we describe the data-driven approaches proposed for identifying and analyzing waiting times (RQ1), followed by the different sources of discovered waiting times (RQ2). Next, we present the results concerning how waiting times can be quantified, i.e., the notion of waiting times (RQ3). Finally, we explain how existing techniques have been validated (RQ4). The comprehensive analysis of the results is provided as supplementary material.<sup>3</sup>

<sup>2</sup> <https://doi.org/10.5281/zenodo.10698027>.

<sup>3</sup> <https://doi.org/10.5281/zenodo.10698027>.



**Table 3** Data Extraction Form

| Meta Data                   | <i>Id</i>                                                                            | Unique Paper Identifier                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Driven Analysis-(RQ1)  | <i>Document Title</i>                                                                | Title of the Paper                                                                                                                                                                                 |
|                             | <i>Authors</i>                                                                       | Authors of the Paper                                                                                                                                                                               |
|                             | <i>Publication Year</i>                                                              | Year of Publication of the Paper                                                                                                                                                                   |
|                             | <i>Citation Count</i>                                                                | Number of Citations Received                                                                                                                                                                       |
|                             | <i>Data Driven Technique (Identification or Analysis)</i>                            | Approach utilized to identify or analyze waiting times in business processes                                                                                                                       |
| Sources Identified-(RQ2)    | <i>Objective (Predictive &amp; What-if, As-Is, Data Quality/ Missing Timestamps)</i> | Analysis of Waiting Times: can be an “as-is” overview of the current situation, a “predictive & What-If” analysis, or a refinement of an event log addressing quality issues or missing timestamps |
|                             | <i>Source of Waiting Time Identified</i>                                             | Sources responsible for waiting times in business processes                                                                                                                                        |
| Quantification Method-(RQ3) | <i>Internal/ External</i>                                                            | Association with the factors affecting the business process, either internal or external                                                                                                           |
|                             | <i>Waiting Times Quantification</i>                                                  | Different notions used for the quantification of waiting times in business processes                                                                                                               |
| Validation Method-(RQ4)     | <i>Approach</i>                                                                      | Validation approaches used for analyzing waiting times in business processes                                                                                                                       |

**Table 4** Paper Selection Process

| Filtering Stage                     | Papers Elimination/<br>Addition by Stage | Total Number of<br>Papers Left |
|-------------------------------------|------------------------------------------|--------------------------------|
| Primary Search Results              | 2611                                     |                                |
| Filtering by paper format           | 173                                      | 2438                           |
| Filtering out duplicates            | 253                                      | 2185                           |
| Filtering by Paper title & abstract | 1950                                     | 235                            |
| Filtering by Reading the Full Paper | 130                                      | 105                            |
| After Backward Referencing          | (105 + 7)                                | 112                            |

#### 4.1 Distribution of Papers by Publication Year

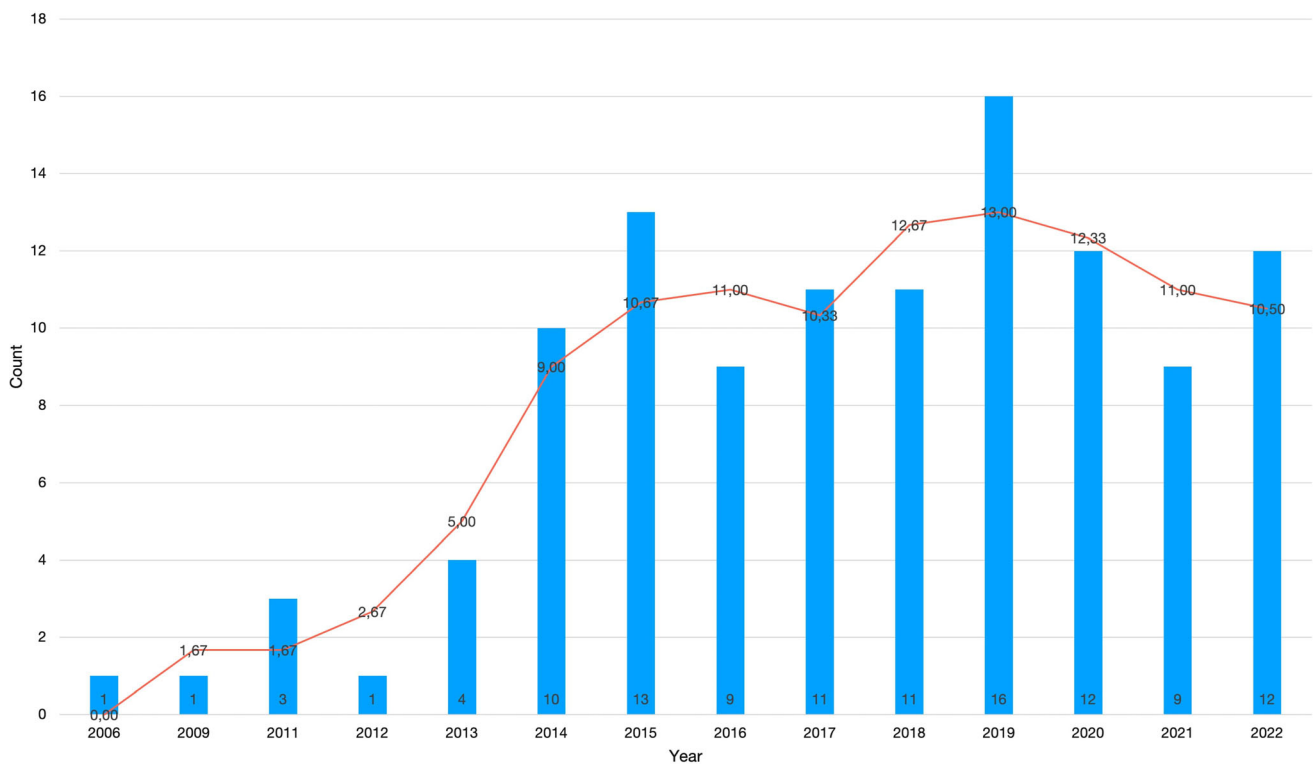
The distribution of relevant papers by year of publication is depicted in Fig. 1. Among the identified papers, we note that the first one was published in 2006. The graph shows that the number of papers published on this topic has increased over time, with more papers published in the last five years than in any other five-year period. The three-year average line shows that the average number of papers published per year has also increased over time. However, we also note that, since 2014, no clear trend can be discerned.

#### 4.2 Data-Driven Techniques for Waiting Times Identification and Analysis

Our review shows that data-driven approaches for identifying and analyzing waiting times can be categorized in three distinct groups: *Descriptive Approaches*, *Predictive & What-If Approaches*, and *Data Refinement Approaches*. Descriptive approaches extract statistics and visualizes business processes to analyze waiting times. For instance, Singh et al. (2021), Drosouli et al. (2020) provides a graph-based visualization of a process using enhanced process maps. Predictive & What-If approaches study the impact of business process changes on the temporal performance of a process. For instance, in Antunes et al. (2019), staff schedules are altered to optimize the temporal performance of a process. Finally, Data Refinement Approaches analyze waiting times to refine event logs and improve data quality. For instance, Goel et al. (2022) propose a quality informed visual miner to assess the accuracy and precision of timestamps in an event log. We explore each of these categories more in-depth in the subsequent sections.

##### 4.2.1 Descriptive Approaches

A subset of the identified approaches use event logs to extract descriptive statistics and create visualizations to help analysts identify and analyze waiting times in an “as-is” process (see Table 5). These approaches have two components. The first involves observing and measuring the waiting times from an event log. The second focuses on



**Fig. 1** Distribution of Papers by Publication Year (Up to August 2022)

aggregating and visualizing the waiting times. In the subsequent discussion, we focus on the latter aspect, namely aggregating and visualizing. A discussion of the former aspect (observing and measuring) can be found in Sect. 4.4.

The bulk of descriptive approaches provide graph-based visualizations of waiting times using Enhanced Process Maps (Singh et al. 2021; Drosouli et al. 2020), Enhanced BPMN Models (Aissaoui et al. 2022), Enhanced Petri Nets (Ganesha et al. 2017a), Statecharts (Leemans et al. 2018), or Temporal Network Representations (Senderovich et al. 2019).

Other studies use chart-based visualization techniques, such as Dotted Charts (Drosouli et al. 2020) or Performance Spectrum Charts (Denisov et al. 2018). Whereas graph-based approaches focus on pinpointing the transitions between activities where waiting times are most prominent, chart-based approaches focus on visualizing how waiting times are distributed across cases, activities, or resources, and on how waiting times evolve over the time-frame of the log.

Another subset of the studies combines graph-based visualizations with Lean Management Tools to provide integrated analysis and redesign approaches to reduce waiting time. For example, Aissaoui et al. (2022) combine enhanced BPMN models with a Lean technique, known as value stream mapping, to analyze wastes, with an emphasis on waiting time. Similarly, Singh et al. (2021) use the

PDSA (Plan, Do, Study, Act) cycle of Lean management together with enhanced process maps to identify bottlenecks in a hospital case.

Two of the identified studies use the Program Evaluation Review Technique (PERT), and Critical Path Method (CPM) to highlight pathways in a process where cycle time could be optimized by reducing waiting times (dos Santos et al. 2020; Rahardianto et al. 2018).

Finally, another set of studies combines machine learning and process mining approaches for analysis of waiting time in “as is” processes. Specifically, Gerhardt et al. (2018) apply association rule mining to find frequent patterns and associations in the event log, while Tri-dalestari et al. (2022) combine clustering with the Enhanced Petri nets to analyze e-commerce processes.

#### 4.2.2 Predictive and What-if Analysis Approaches

Another complementary set of studies focuses on predicting the future states of a process. Here, we can categorize the approaches into two distinct groups (see Table 6). The first group aims to predict the remaining time of ongoing cases under the assumption that no changes are made to the process. These approaches are here called “Predictive Approaches”. Another subset of approaches focuses on predicting the temporal performance of the process, assuming that a change, such as adding resources, is made.

**Table 5** Descriptive Analytical Approaches (As-Is Process Analysis)

| Category         | Technique (identification or Analysis)                  | Objective                           | Article References                                                                                                                                                                                                                                                                                                                            |
|------------------|---------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graph            | Enhanced BPMN Model                                     | As-is Waiting Time Analysis         | Aissaoui et al. (2022)                                                                                                                                                                                                                                                                                                                        |
| Graph            | Enhanced Petri-nets                                     | As-is Waiting Time Analysis         | Ganesha et al. (2017a), Tridalestari et al. (2022), Ganesha et al. (2017b), Broderick et al. (2011), Satitcharoenmuang et al. (2017), Rahardianto et al. (2018)                                                                                                                                                                               |
| Graph            | Enhanced Process Map                                    | As-is Waiting Time Analysis         | Singh et al. (2021), Drosouli et al. (2020), Yampaka and Chongstitvatana (2016), Ganesha et al. (2017a), Gerhardt et al. (2018), Rojas et al. (2018), Duong et al. (2021), Pan and Zhang (2021), Porouhan and Premchaiswadi (2018), Satitcharoenmuang et al. (2017), Senderovich et al. (2019), dos Santos et al. (2020), Budai et al. (2019) |
| Graph            | Temporal Network Representation                         | As-is Waiting Time Analysis         | (Senderovich et al. (2019)                                                                                                                                                                                                                                                                                                                    |
| Graph            | Statechart                                              | As-is Waiting Time Analysis         | (Leemans et al. (2018)                                                                                                                                                                                                                                                                                                                        |
| Chart            | Dotted Chart Analysis                                   | As-is Waiting Time Analysis         | (Drosouli et al. (2020), Pan and Zhang (2021)                                                                                                                                                                                                                                                                                                 |
| Chart            | Performance Spectrum                                    | As-is Waiting Time Analysis         | (Denisov et al. (2018, Klijn and Fahland (2019)                                                                                                                                                                                                                                                                                               |
| Machine Learning | Association Rule Mining                                 | As-is Waiting Time Analysis         | (Gerhardt et al. (2018)                                                                                                                                                                                                                                                                                                                       |
| Machine Learning | K-Mean Clustering                                       | As-is Waiting Time Analysis         | (Tridalestari et al. (2022)                                                                                                                                                                                                                                                                                                                   |
| Other            | Davies Bouldin Approach                                 | As-is Waiting Time Analysis         | (Tridalestari et al. (2022)                                                                                                                                                                                                                                                                                                                   |
| Other            | Hierarchical Timed Petri-net Model (Coloured Petri-net) | As-is Waiting Time Analysis         | (Salimifard et al. (2013)                                                                                                                                                                                                                                                                                                                     |
| Other            | Semantic-based multilevel Abstraction Mechanism         | As-is Waiting Time Analysis         | (Leonardi et al. (2018)                                                                                                                                                                                                                                                                                                                       |
| Other            | PERT/CPM                                                | Identifying Critical Paths          | (dos Santos et al. (2020)                                                                                                                                                                                                                                                                                                                     |
| Other            | Generalized Network Problem Method                      | As-is Waiting Time Analysis         | (Budai et al. (2019)                                                                                                                                                                                                                                                                                                                          |
| Other            | PERT                                                    | Identifying Critical Paths          | (Rahardianto et al. (2018)                                                                                                                                                                                                                                                                                                                    |
| Other            | ARPRA                                                   | As-is Waiting Time Analysis         | (Martin et al. (2015)                                                                                                                                                                                                                                                                                                                         |
| Other            | Value Stream Mapping                                    | Identifying non-value-adding wastes | (Aissaoui et al. (2022)                                                                                                                                                                                                                                                                                                                       |
| Other            | PDSA (Lean Six Sigma)                                   | Reduce Waiting Times                | (Singh et al. (2021)                                                                                                                                                                                                                                                                                                                          |
| Other            | Context Aware Performance Analysis                      | As-is Waiting Time Analysis         | (Hompes et al. (2016)                                                                                                                                                                                                                                                                                                                         |

These approaches are referred to as “What-If Analysis Approaches”. We excluded approaches that primarily focus on predicting case outcomes, such as Teinemaa et al. (2019). Moreover, we included some works from the area of predictive process monitoring if they explicitly seek to identify or analyze waiting times in a business process.

The approaches under what-if analysis can be further classified according to the way they represent a business process and its temporal performance. The bulk of the methods rely on graph-based representations, including BPMN Models (Dijkman et al. 2018), Process Maps (Antunes et al. 2019), and Petri nets (Nguyen et al. 2016). In Antunes et al. (2019), the authors propose a framework



that combines enhanced process maps, mixed integer programming, and discrete event simulation to optimize resource utilization and waiting times by altering the staff schedules. Similarly, Low et al. (2014) proposed a genetic algorithm to explore a search space of possible re-design options in order to optimize resource utilization and waiting times. On the other hand, Pika et al. (2013) propose an optimization technique to minimize the probability of delays.

Another set of studies employ machine learning techniques to address issues related to waiting times in business processes. Techniques such as Support Vector Machines, Logistic Regression, Naive Bayes, and Decision Trees are used to predict the likelihood of a case finishing either on time or with a delay (Khan et al. 2019; Park and Song 2020). Typically, these methods incorporate information about waiting times within a process into the feature selection process for training machine learning models aimed at predicting delays in business processes.

Finally, Mannhardt et al. (2019) present a method based on Markov Chain Monte Carlo sampling (MCMC) to predict delays in a railway incident management process, while Senderovich et al. (2016) use Generalized Stochastic Petri nets (GSPNs) to predict sojourn times for each activity of a process. Moreover, there are a handful of studies that not only identify waiting times but also link them to actionable improvement opportunities to reduce these waiting times (van der Aalst et al. 2015).

#### 4.2.3 Data Refinement Approaches

Several studies have explored analyzing waiting times to address specific gaps in event logs. To address data quality issues such as missing start time stamps, a set of papers have focused on analyzing waiting times. For instance, when some or all start time stamps are missing in an event log, having knowledge of the usual waiting time between activities aids in making estimates for the missing

**Table 6** Predictive Approaches

| Category         | Technique (identification or Analysis)               | Objective                     | Article References                                                   |
|------------------|------------------------------------------------------|-------------------------------|----------------------------------------------------------------------|
| Simulation       | Discrete Event Simulation                            | What-if Waiting Time Analysis | Antunes et al. (2019), Cho et al. (2019)                             |
| Simulation       | Queue Modeling                                       | What-if Waiting Time Analysis | Dijkman et al. (2018)                                                |
| Simulation       | Queue Mining                                         | What-if Waiting Time Analysis | Senderovich et al. (2015)                                            |
| Graph            | Enhanced Process Maps                                | What-if Waiting Time Analysis | Antunes et al. (2019), Cho et al. (2019)                             |
| Graph            | Enhanced Petri-nets                                  | What-if Waiting Time Analysis | (Nguyen et al. (2016), Senderovich et al. (2015), Zeng et al. (2020) |
| Graph            | Stochastic Petri-nets                                | What-if Waiting Time Analysis | (Rogge-Solti and Weske (2015)                                        |
| Graph            | Stochastic Petri-net                                 | Predict Sojourn Times         | (Senderovich et al. (2016)                                           |
| Graph            | Resource Aware Petri-Net                             | Predict Delays                | (Pla et al. (2011)                                                   |
| Graph            | Annotated Process Model                              | Predict Delays                | (Park and Song (2020)                                                |
| Graph            | Enhanced BPMN Model                                  | What-if Waiting Time Analysis | (Dijkman et al. (2018), Grabis (2014)                                |
| Optimization     | Genetic Algorithm                                    | What-if Waiting Time Analysis | (Low et al. (2014)                                                   |
| Optimization     | Mixed Integer Programming                            | What-if Waiting Time Analysis | (Antunes et al. (2019)                                               |
| Machine Learning | Deep Learning (CNN, LSTM, and LRCN)                  | Predict Delays                | (Park and Song (2020)                                                |
| Machine Learning | SVM, Logistic Regression, Naïve Bayes, Decision Tree | Predict Delays                | (Khan et al. (2019)                                                  |
| Other            | Process Risk Indicator                               | Predict Delays                | (Pika et al. (2013)                                                  |
| Other            | Markov chain Monte Carlo sampling (MCMC)             | Predict Delays                | (Mannhardt et al. (2019)                                             |
| Other            | Custom Approach                                      | What-if waiting time analysis | Pufahl et al. (2014)                                                 |

timestamps. For instance, in Fracca et al. (2022), they derive an enhanced BPMN model from an event log and employ simulation techniques to estimate these missing start time stamps. In Goel et al. (2022), the authors present the Quality-Informed Visual Miner that uses a probabilistic method, informed by waiting times, to assess the accuracy and precision of timestamps in an event log. Taking a statistical approach, Nogayama and Takahashi (2015) apply Expectation and Maximization Techniques. They estimate average latent waiting and service times based on Maximum Likelihood Estimation (MLE). Here, the waiting times serve as foundational metrics to enhance the event log. Lastly, Thomas et al. (2015) introduced the Alf Miner technique. By analyzing the behavior of resources in a process and examining waiting times of activities, they determine the next probable activity (NPA) and the resource assigned to perform it. This approach, while focusing on resource behavior, underscores waiting times as a determinant in predicting future activities. We categorize this technique as data refinement since it leverages waiting times to extrapolate information about incomplete instances in an event log. In essence, all these studies, while addressing diverse challenges, converge on the importance of waiting times as a metric and tool for refining and enhancing the quality of event logs (Table 7).

### 4.3 Sources of Waiting Time

With respect to the causes (herein called *sources*) of waiting time, our review shows that the papers can be classified into two groups: those that consider external sources of waiting time and those that consider internal sources (see Table 8). By external sources of waiting times, we refer to causes that are beyond what is captured in an event log. While some of these causes are external to the event log or even external to the process itself, they might not always be external to the organization where the process is executed. For instance, external causes can be waiting for a delivery from a supplier; waiting for a response from a customer (Ferreira and Vasilyev 2015); waiting for the recovery of a patient after a treatment (Jaisook and Premchaiswadi 2015; Mans et al. 2012); or waiting times due to exceptions or incidents not recorded in the event log (Mannhardt and Landmark 2019). Note that external waiting times could be due to resources in the process being busy on activities not recorded in the event log, whether these are activities in the same process described by the event log, or activities belonging to other processes. However, none of the reviewed studies considered waiting times stemming from unrecorded activities.

Internal sources of waiting time, on the other hand, are waiting times caused by factors that are recorded in the input event log. In the studies we identified, we found four

types of internal waiting times; (1) due to resource contention; (2) due to resource unavailability; (3) due to prioritization; and (4) due to batching. Resource contention is a phenomenon that occurs when an activity instance is *enabled* (i.e., ready to be executed), but not started because the resource(s) that would normally execute it is busy performing another activity instance (Petitdemange et al. 2020; Antunes et al. 2019). Waiting times due to resource unavailability occur when an activity instance is enabled, but it cannot be started because the resource(s) who would normally perform it is off duty (Aissaoui et al. 2022; Pang et al. 2021). Waiting time due to prioritization occurs when an activity instance is enabled, a resource that could perform it is available, but said resource starts another activity instance with higher priority, even though this latter activity instance became enabled after the former (lower priority) activity instance (Cho et al. 2019; Rojas et al. 2018; Premchaiswadi and Porouhan 2015). Finally, waiting times due to batching occurs when an activity instance is enabled, but it is not started because the resource who eventually performs it, bundles it together with other activity instances and performs all the activity instances in a single go (Lashkevich et al. 2022).

### 4.4 Notions of Waiting Time

In addressing RQ3, we examined the relevant corpus of papers to identify how waiting times are quantified. Our review reveals that there are different notions of waiting times which are employed when quantifying them in business processes. First, we found that some studies define waiting time at the granularity of a case (*case waiting time*), while others do so at the granularity of each activity instance (*activity waiting time*).

Second, irrespective of the granularity, we identified five distinct approaches to measure waiting time: (1) simple waiting time; (2) control-flow waiting time; (3) idle time; (4) shelf time; and (5) customer perceived idle time (see Table 9). Below, we review each of these five measurement approaches and relate them to the two levels of granularity. In this review, we use an example of a business process depicted in BPMN notation (see Fig. 2) to illustrate the notions of waiting times. Here, every case initiates with Activity A followed by a parallel block of activities B and C. The process reaches completion following this parallel block. For specificity, we are considering three executions of this process: Case1, Case2, and Case3. These cases involve three resources, X, Y, and Z, as illustrated in Fig. 3.

Simple waiting time is defined as the start time of an activity instance minus the end time of the most recently completed activity instance of the same case. For example, with reference to Fig. 3, the time between the end of activity A in case 1 and the start of activity B in case 1 is a

**Table 7** Data Refinement Approaches

| Category                            | Technique (identification or Analysis)  | Objective                                    | Article References            |
|-------------------------------------|-----------------------------------------|----------------------------------------------|-------------------------------|
| Graph and Simulation Based Approach | Enhanced BPMN model and simulation      | Estimate missing start timestamps            | Fracca et al. (2022)          |
| Graph Based Approach                | AlfyMiner                               | Estimate next probable activity and resource | Thomas et al. (2015)          |
| Statistical Approach                | Expectation and Maximization Algorithms | Estimate waiting and service times           | Nogayama and Takahashi (2015) |
| Probabilistic Approach              | Quality Informed Visual Miner           | Assess accuracy/precision of timestamps      | Goel et al. (2022)            |

**Table 8** Sources of Waiting Times Identified

| Source of Waiting time identified      | Internal/ External | Article References                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Contention                    | Internal           | Pan and Zhang (2021), Denisov et al. (2018), Antunes et al. (2019), Singh et al. (2021), Hompes et al. (2016), Ganesha et al. (2017a), Cho et al. (2019), Gerhardt et al. (2018), Tridalestari et al. (2022), Fracca et al. (2022), Drosouli et al. (2020), Senderovich et al. (2015), Zeng et al. (2020), Thomas et al. (2015), Salimifard et al. (2013), Budai et al. (2019), Low et al. (2014), Pla et al. (2011) |
| Resource Unavailability                | Internal           | Mannhardt et al. (2019), Pika et al. (2013, Aissaoui et al. (2022)                                                                                                                                                                                                                                                                                                                                                   |
| Case Prioritization                    | Internal           | Rojas et al. (2018, Cho et al. 2019)                                                                                                                                                                                                                                                                                                                                                                                 |
| Batching                               | Internal           | Lashkevich et al. (2022, Duong et al. (2021, Pufahl et al. (2014, Klijn and Fahland (2019)                                                                                                                                                                                                                                                                                                                           |
| Patients Waiting for the Appointments  | Internal           | Aissaoui et al. (2022)                                                                                                                                                                                                                                                                                                                                                                                               |
| Malfunctioning of Support System       | External           | Mannhardt et al. (2019)                                                                                                                                                                                                                                                                                                                                                                                              |
| Waiting for a Response from a Customer | External           | Ferreira and Vasilyev (2015)                                                                                                                                                                                                                                                                                                                                                                                         |
| Recovery of a Patient                  | External           | Jaisook and Premchaiswadi (2015, Mans et al. (2012)                                                                                                                                                                                                                                                                                                                                                                  |
| Delay Due to Incidents                 | External           | Mannhardt and Landmark (2019)                                                                                                                                                                                                                                                                                                                                                                                        |

simple waiting time. The same holds for the time between the end of activity B in case 1 and the start of activity C in case 1. Most of the reviewed studies (77 studies) use this notion of simple waiting time notion to quantify the waiting time at the activity instance level (Petitdemange et al. 2020). We note that simple waiting time is defined at the level of each activity instance. The simple waiting time of a case is the sum of the waiting times of the activity instances in the case.

Control-flow waiting time is the start time of an activity instance minus the enablement time of that activity instance. Here, the enablement time of an activity instance is the time when the activity instance was enabled from a control-flow perspective. Let us assume, for example, that activities B and C are parallel activities, meaning that they can be executed in any order or their execution may overlap. Under this assumption, the control-flow waiting

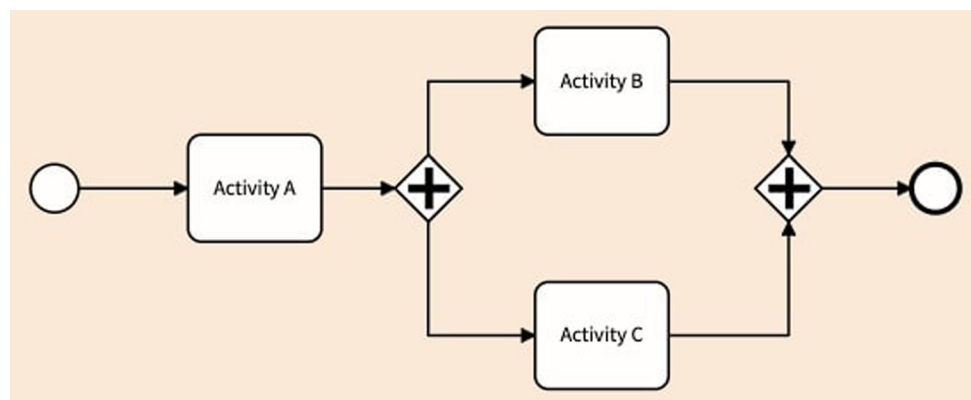
time of Activity C in case 1 is the start time of this activity instance and the end time of activity A in case 1, as shown in Fig. 3. Note that the control-flow waiting time of Activity C in case 1 is not the same as its simple waiting time. On the other time, the control-flow waiting time of activity instance C is equal to its simple waiting time, because the activity that immediately precedes C in case 1 is A, and this latter is the activity that enables C.

In Fracca et al. (2022), the authors use the notion of control flow waiting time to estimate missing start timestamps of activity instances in a process. Again, we note that control-flow waiting time is defined at the granularity of an activity instance. The control-flow waiting time of a case may be calculated by summing those of its activity instances.

Idle time is a case-level notion of waiting time. The idle time of a case is the sum of the time periods during which

**Table 9** Different Notions of Waiting Times

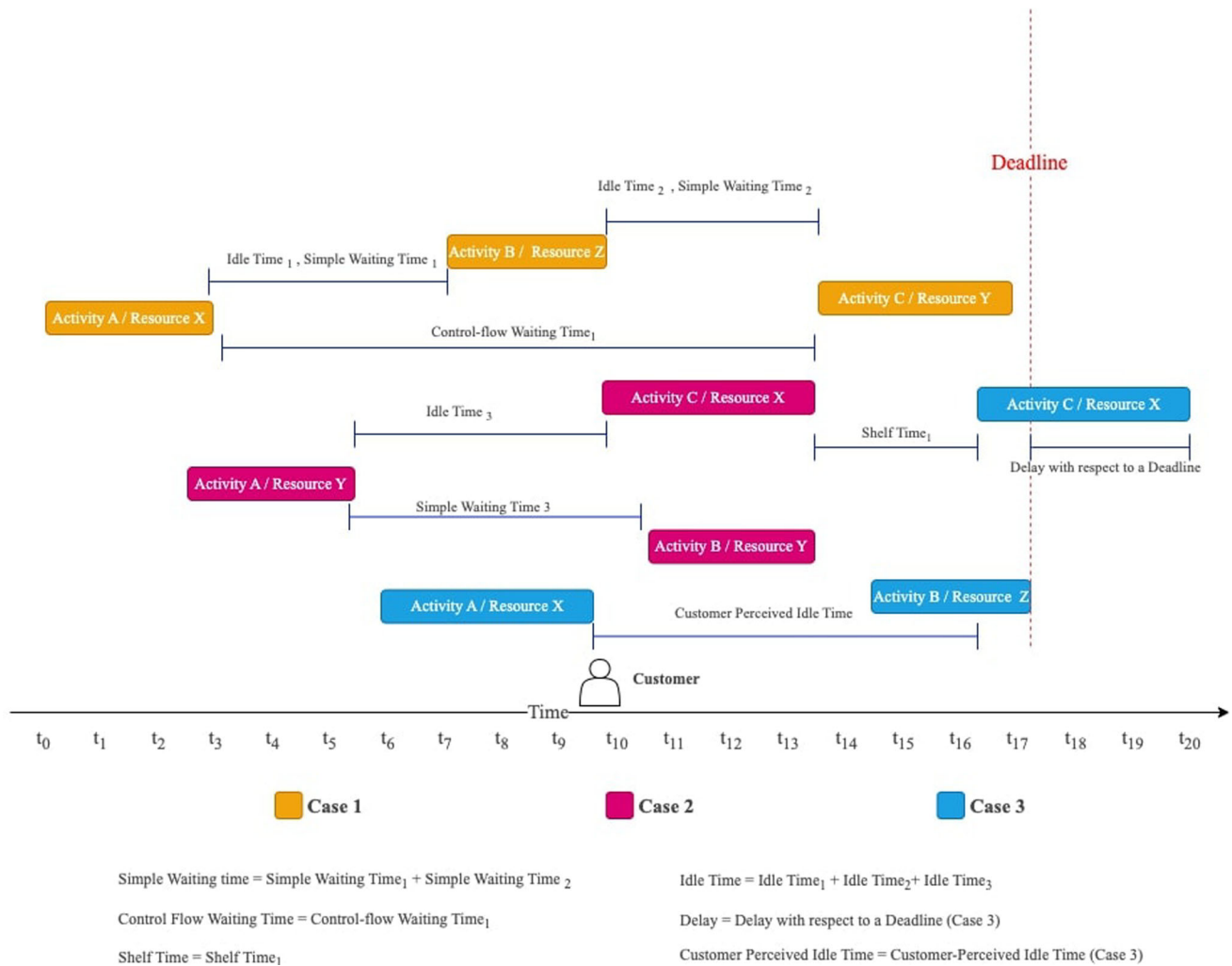
| Notions of Waiting Time          | Article References                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simple Waiting Time              | Aissaoui et al. (2022), Singh et al. (2021), Hompes et al. (2016), Rojas et al. (2018), Ganesha et al. (2017b), Broderick et al. (2011), Pufahl et al. (2014), Porouhan and Premchaiswadi (2018), Drosouli et al. (2020), Antunes et al. (2019), Duong et al. (2021), Yampaka and Chongstitvatana (2016), Ganesha et al. (2017a), Cho et al. (2019), Gerhardt et al. (2018), Pan and Zhang (2021), Satitcharoenmuang et al. (2017), Fracca et al. (2022), Tridalestari et al. (2022), Grabis (2014), Nguyen et al. (2016), Rahardianto et al. (2018), Thomas et al. (2015), Nogayama and Takahashi (2015), Salimifard et al. (2013), Leonardi et al. (2018), dos Santos et al. (2020), Budai et al. (2019), Senderovich et al. (2016), Rogge-Solti and Weske (2015), Denisov et al. (2018), Low et al. (2014), Park and Song (2020), Goel et al. (2022), Martin et al. (2015) |
| Control-Flow Waiting Time        | Senderovich et al. (2019), Leemans et al. (2018), Senderovich et al. (2016), Pika et al. (2013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Idle Time                        | Drosouli et al. (2020), dos Santos et al. (2020), Denisov et al. (2018), Klijn and Fahland (2019), Low et al. (2014)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Shelf Time                       | Andrews and Wynn (2017)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Delay With Respect to a Deadline | Khan et al. (2019), Ganesha et al. (2017a), Gerhardt et al. (2018), Pan and Zhang (2021), Satitcharoenmuang et al. (2017), Fracca et al. (2022), Senderovich et al. (2015), Zeng et al. (2020), Thomas et al. (2015), Senderovich et al. (2019), Mannhardt et al. (2019), dos Santos et al. (2020), Rogge-Solti and Weske (2015), Low et al. (2014), Pla et al. (2011), Pika et al. (2013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Customer Perceived Idle Time     | Aissaoui et al. (2022), Singh et al. (2021), Antunes et al. (2019), Yampaka and Chongstitvatana (2016), Ganesha et al. (2017a), Cho et al. (2019), Tridalestari et al. (2022), Grabis (2014), Salimifard et al. (2013), Rogge-Solti and Weske (2015)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Fig. 2** Example Business Process BPMN Notation

no work is being done. Figure 3 shows that case 1 has two idle time periods; one between end of Activity A and start of B, and another between the end of B and start of C. In contrast, Case 2 only has one idle time period; the one between the end of Activity A and the start of Activity C. There is no idle time between Activity B and C because activity B is performed during the execution of C. Idle time is typically used as a measure of temporal efficiency. For example, Drosouli et al. (2020) propose to measure the efficiency of customer-facing processes, with reference to a bike rental process using the idle time(s). In scenarios where there are no overlaps in activity times and no other time gaps apart from the one between two consecutive activities, the sum of simple waiting times for all activities would equal the total idle time. However, if there are other factors such as parallel activities, external delays, or other

non-activity related idle times, then the sum of the simple waiting times might not necessarily equate to the total idle time.

Shelf time (Andrews and Wynn 2017) is the amount of time that occurs when an activity instance is enabled, the resource who ultimately performs this activity instance is also available, but the activity instance is not started due to external factors (e.g., a resource waiting for feedback from a customer). In other words, shelf time is a measure of external waiting time. With reference to Fig. 3, the time between the completion of Activity C in Case 2 and the start of Activity C in Case 3 is a shelf time because the resource that executed Activity C in Case 3 (Resource X) became available after the completion of Activity C in Case 2. Yet, resource X did not start Activity C in Case 3, even though this latter activity instance was enabled. Note



**Fig. 3** Different Notions of Waiting Times in Business Processes

that shelf time is defined at the activity instance level. It can be lifted to the case level by summing the shelf times of the activity instances in a case.

Not every waiting time is relevant from a business process optimization perspective. For example, in a customer-facing process, such as a patient treatment process, waiting times that are not perceived as such by the patient are not relevant for process optimization. Only waiting times that are perceived as such by the customer are relevant (Abo-Hamad 2017; Rojas et al. 2018). A *touch-point*, in this context, is defined as an activity that involves direct engagement with the customer. A subset of studies (19 papers) employ the notion of *customer-perceived idle time*. This concept denotes the span between two customer touch-points. The idea behind customer-perceived idle time is that it represents a subset of the overall idle time for a case. It captures the intervals specifically observed by the customer, especially when some activities in the process qualify as touch-points because of their direct interaction

with the customers. The idle time that occurs between the end of one touch-point and the beginning of another, within the same case, is defined as customer-perceived idle time. In contrast, idle times before the first or after the last touch-point are not categorized as customer-perceived idle time. As shown in Fig. 3, before the start of Activity C for Case 3 and the end of Activity A for Case 3, the customer is waiting to be served by a resource between two touch-points. This is an example of customer-perceived idle time.

We note that some studies do not seek to measure waiting time per se, but rather the delay of a case with respect to a deadline, e.g., Gupta et al. (2014). For example, in Fig. 3, Activity C of Case 3 was completed after the deadline. Hence, a delay with respect to a deadline refers to the amount of time that elapses between the deadline for an activity and the actual time it is completed. We classify this type of delay (with respect to a deadline) as a subset of a customer-perceived idle time. From a customer's perspective, the waiting time begins the moment a deadline is



missed. Once an agreed-upon deadline is not met, customers often perceive any additional waiting time as an unnecessary delay. In this context, the idle period for the customer starts the moment the deadline passes without the activity being completed. Thus, it can be seen as a specific instance where the customer perceives a lapse in service or a halt in progress.

#### 4.5 Validation Methodology

The reviewed studies also differ in terms of the approach employed to demonstrate the applicability, usability, and/or benefits of the methods they propose (see Table 10). A subset of studies, such as Dijkman et al. (2018), Broderick et al. (2011), rely on an illustrative example to support their claims. Such illustrative examples take the form of a step-by-step application of the method on a real-world example, an application of the method on a hypothetical scenario, or using a synthetic log. In either case, the application of the method is conducted by the authors of the study without external validation.

Other studies validate the proposed method(s) using case studies, such as Aissaoui et al. (2022), Drosouli et al. (2020), Antunes et al. (2019). The use of case studies as a validation approach offers several advantages. Firstly, case studies provide an in-depth and contextual analysis, allowing practitioners to understand the practical applicability of the proposed methods in real-world settings. Case studies can reveal unforeseen challenges and benefits that might not be evident in controlled experiments or simulations. Secondly, they facilitate direct engagement with actual users, which enhances the ecological validity of the findings. By incorporating feedback from these end users, the proposed methods can be refined and optimized for practical implementation. Hence, the integration of case studies as a validation mechanism ensures that the method is not only theoretically sound but also practically effective.

Other studies use computational experiments to validate the goodness of the proposed method. In this category, we find studies that propose simulation or predictive modeling methods (Rogge-Solti and Weske 2015) and data refinement methods (Goel et al. 2022) where there is a clear ground-truth against which the output of the proposed method can be compared. A subset of these studies, such as Senderovich et al. (2019), include a comparison between the proposed method and some baseline approach, with respect to the ground truth.

## 5 Discussion

In this section, we examine research gaps, implications, and validity threats identified by our review, highlighting limitations of current waiting time analysis approaches and suggesting avenues for future research.

### 5.1 Research Gaps and Implications

#### 5.1.1 Lack of Actionable Recommendations

We noted a lack of *actionability* in the existing corpus of research work. Actionability refers to recommendations that enable the process owners or other relevant stakeholders to reduce the waiting times, while considering other relevant performance dimensions of a business process, such as costs and quality. As a representative example, (Andrews and Wynn 2017) provide a detailed approach for decorticating shelf time in a business process. However, this study does not provide guidelines or recommendations on interventions or changes in the process that could help to reduce shelf times. Similarly, predictive approaches, such as (Khan et al. 2019; Park and Song 2020) focus on making predictions about remaining waiting time or cycle time, but they do not link these predictions to concrete recommendations that would enable process workers to minimize waiting times. Thus, a research gap and, thereby, an avenue for future work would be developing methods that build on descriptive and predictive approaches and link their results to actions that optimize waiting times of a process.

#### 5.1.2 Need for Multi-Causal Analysis Approaches

Our review shows that existing approaches adopts a mono-source (i.e., mono-causal) approach, meaning that they focuses on one source (or cause) of waiting time in isolation. However, there are clear dependencies between the different sources of waiting time in a process. For example, if resource contention is reduced by deploying additional resources or by reducing workload via automation, this can alleviate the waiting time attributable to resource contention. However, if the same process contains a batched activity that is executed once per week, the effects of reducing waiting times due to resource contention may be nullified by the presence of the weekly batch (i.e., the waiting time would still remain in the order of a few days due to the batching rule). This example illustrates the need for multi-causal approaches to waiting time analysis, which we highlight as another gap in the literature.

**Table 10** Validation Methodology

| Validation Approach      | Article Reference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case Study               | Aissaoui et al. (2022), Drosouli et al. (2020), Pan and Zhang (2021), Pufahl et al. (2014), Satitcharoenmuang et al. (2017), Gerhardt et al. (2018), Salimifard et al. (2013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Computational experiment | Antunes et al. (2019), Grabis (2014), Cho et al. (2019), Senderovich et al. (2015), Thomas et al. (2015), Senderovich et al. (2016), Low et al. (2014), Pla et al. (2011), Fracca et al. (2022), Senderovich et al. (2019), Mannhardt et al. (2019), Rogge-Solti and Weske (2015), Park and Song (2020)                                                                                                                                                                                                                                                                                                                                                 |
| Illustrative Example     | Dijkman et al. (2018), Broderick et al. (2011), Yampaka and Chongstitvatana (2016), Ganesha et al. (2017a, b), Tridalestari et al. (2022), Rojas et al. (2018), Duong et al. (2021), Fracca et al. (2022), Singh et al. (2021), Porouhan and Premchaiswadi (2018), Hompes et al. (2016), Khan et al. (2019), Nguyen et al. (2016), Rahardianto et al. (2018), Zeng et al. (2020), Senderovich et al. (2019), Nogayama and Takahashi (2015), Leemans et al. (2018), Leonardi et al. (2018), dos Santos et al. (2020), Budai et al. (2019), Denisov et al. (2018), Klijn and Fahland (2019), Pika et al. (2013), Goel et al. (2022), Martin et al. (2015) |

### 5.1.3 External Causes of Waiting Times

In the studies reviewed in this SLR, there is a strong emphasis on internal causes of waiting times, whereas discussions on different types of external waiting times is limited. Arguably, to decompose external waiting times into multiple causes we would require additional metadata or information not traditionally captured in an event log. This is another gap and potential direction for future work.

### 5.1.4 Improving Process Analysis

In our literature review, we observed an even distribution between studies on descriptive analysis and predictive or ‘what-if’ analysis approaches, with only a handful of approaches delving into data refinement methodologies. Descriptive approaches presuppose the existence of start and end times of a case in an event log, enabling the identification and analysis of waiting times in business processes. On the other hand, ‘predictive & what-if’ approaches do not necessarily operate under the assumption of having both timestamps in an event log. These approaches integrate reasoning about waiting times and processing times using complex feature sets, which may include information about resources and workload, such as the number of active cases at a given point in time. Consequently, these approaches incorporate information about waiting times indirectly without providing explicit representations. This gap suggests an opportunity for future research to develop methodologies that not only bridge the divide between descriptive and predictive analyses but also enhance the representation and understanding of waiting times through advanced data refinement techniques. By leveraging the strengths of both approaches, such research could provide deeper insights into process efficiencies, resource allocation, and overall business process improvement.

### 5.1.5 Measuring Waiting Times with Enablement Time

Another research gap relates to the problem of quantifying waiting times in business processes. The majority of approaches, with the exception of a few such as (Lashkevich et al. 2022), use simple waiting time. However, they do not consider the enablement time, i.e., time when activity instances become ready to be executed. Including enablement time into the calculation of waiting time can improve the precision of waiting time. The challenge here is how to infer the enablement time of activities in the absence of explicit indicators of activity instance enablement in an event log. Accordingly, we see an opportunity for developing methods to reliably infer activity enablement times from event logs, and using this information for accurate waiting time quantification.

### 5.1.6 Lack of Ecological Validity and External Validation

Our review also shows that a share of studies rely on illustrative examples to support their claims of applicability or usability. This approach provides limited (or no) ecological validity to the findings. Other studies rely on computational experiments which allow authors to back claims of accuracy (e.g., in the context of predictive modeling). Likewise, such methods are limited in terms of validating claims of usability in practice. As such, we highlight the need for external validation (e.g., via case studies or usability evaluations).

### 5.1.7 Implications for Practitioners

Our study also points to the practical implications for practitioners. Practitioners would benefit by being careful when using existing approaches for analyzing waiting times in business processes. They should examine the specific definitions of waiting times that these approaches

use. In addition, practitioners benefit from combining current analysis approaches with other approaches to identify candidate improvements to the process. For example, applying data-driven waiting time analysis with techniques from Lean Six Sigma (Graafmans et al. 2021) can lead to the identification of actions that improve business processes. This combination provides a broader approach to improving processes by using a variety of well-established techniques.

#### 5.1.8 Implications for Researchers

Our study also brings along some implications for researchers. First, researchers in the field should carefully elucidate the definition of waiting times as there are varying notions of these. Second, it is beneficial for researchers in process mining to correlate waiting time analysis with its possible causes to enable linking these causes to potential remedial actions. The clarity in defining and applying waiting times can aid in devising more effective and contextually relevant solutions and interventions.

#### 5.2 Threats to Validity

SLR studies are prone to validity threats due to overlooking or excluding relevant publications (Keele et al. 2007; Randolph 2007). We addressed this threat by following the guidelines for development of search strings (Keele et al. 2007) and conducting backward referencing to reduce the risk of overlooking relevant publications. To further minimize the risk of eliminating relevant publications, we followed the guidelines by specifying clear inclusion and exclusion criteria, filtering on the safe side (rather include than exclude), and discussing papers that were not clearly irrelevant. SLR studies also have a limitation of data extraction bias. To reduce this risk, we discussed each paper in the final list and modified the data extraction whenever required.

## 6 Conclusion

This article offers an insight into data-driven approaches for identifying and analyzing waiting times in business processes. The contribution of the paper is a multi-dimensional taxonomy of such approaches in terms of their purpose, sources (causes) of waiting time, notions of waiting times, and their validation approach. It is observed that many proposals in this field aim at descriptively analyzing waiting times in “as-is” processes. Some focus on predicting waiting times or estimating missing timestamps, while others quantify the quality of timestamps in an event

log. A predominant focus is placed on waiting times due to resource contention. We also highlighted discrepancies in the adopted notions of waiting times and a lack of validation in practice, e.g., via case studies.

We have pinpointed four research gaps that offer avenues for further exploration. These are the development of approaches providing actionable recommendations to minimize waiting times, holistic identification of multiple sources of waiting times, incorporation of enablement time to refine waiting time estimates from event logs, and enhanced validation of proposed methods through studies ensuring external validity. These gaps present opportunities to further develop this research area and to extend the exploration of strategies and methods to optimize waiting times in business processes effectively.

**Acknowledgements** This research is supported by the European Research Council (PIX Project).

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

- Abo-Hamad W (2017) Patient pathways discovery and analysis using process mining techniques: an emergency department case study. *Springer proceedings in mathematics & statistics*. Springer International, Cham, pp 209–219
- Aissaoui NO, Mbarek HB, Layeb SB, Hadj-Alouane AB (2022) A BPMN-VSM based process analysis to improve the efficiency of multidisciplinary outpatient clinics. *Production Planning & Control* pp 1–31
- Andrews R, Wynn MT (2017) Shelf time analysis in CTP insurance claims processing. In: *PAKDD (workshops)*, Springer, Heidelberg, LNCS, vol 10526, pp 151–162
- Antunes BBP, Manresa A, Bastos LSL, Marchesi JF, Hamacher S (2019) A solution framework based on process mining, optimization, and discrete-event simulation to improve queue performance in an emergency department. *Business process management workshops*, Springer, Heidelberg, LNBIP 362:583–594
- Augusto A, Conforti R, Dumas M, Rosa ML, Maggi FM, Marrella A, Mecella M, Soo A (2019) Automated discovery of process models from event logs: review and benchmark. *IEEE Trans Knowl Data Eng* 31(4):686–705
- Brereton P, Kitchenham BA, Budgen D, Turner M, Khalil M (2007) Lessons from applying the systematic literature review process

- within the software engineering domain. *J Syst Softw* 80(4):571–583
- Broderick JA, Allen LV, Tilbury DM (2011) Anomaly detection without a pre-existing formal model: application to an industrial manufacturing system. In: CASE, IEEE, pp 169–174
- Budai I, Kocsi B, Pusztai L (2019) New approach for resource allocation in digital healthcare 4.0. In: Proceedings 5th CARPE conference: Horizon europe and beyond, Universitat Politècnica València, pp 244–251
- Capitán-Agudo C, Salas-Urbano M, Cabanillas C, Resinas M (2022) Analyzing how process mining reports answer time performance questions. *BPM*, Springer, Heidelberg, LNCS 13420:234–250
- Cho M, Song M, Yoo S, Reijers HA (2019) An evidence-based decision support framework for clinician medical scheduling. *IEEE Access* 7:15,239–15,249
- Denisov V, Fahland D, van der Aalst WMP (2018) Unbiased, fine-grained description of processes performance from event data. *BPM*, Springer, Heidelberg, LNCS 11080:139–157
- Diba K, Remy S, Pufahl L (2019) Compliance and performance analysis of procurement processes using process mining. In: International conference on process mining
- Dijkman RM, Adan I, Peters S (2018) Advanced queueing models for quantitative business process analysis. In: SEAA, IEEE Computer Society, pp 260–267
- Drosouli I, Theodoropoulou G, Miaoulis G, Voulodimos A (2020) A process mining approach for resource allocation management in a bike sharing system. In: PCI, ACM, pp 327–333
- Dumas M, Rosa ML, Mendling J, Reijers HA (2018) Fundamentals of business process management, vol 2. Springer, Heidelberg
- Dunzer S, Stierle M, Matzner M, Baier S (2019) Conformance checking: a state-of-the-art literature review. In: S-BPM ONE, ACM, pp 4:1–4:10
- Duong LT, Travé-Massuyès L, Subias A, Roa NB (2021) Assessing product quality from the production process logs. *Int J Adv Manuf Technol* 117(5–6):1615–1631
- Ferreira DR, Vasilyev E (2015) Using logical decision trees to discover the cause of process delays from event logs. *Comput Ind* 70:194–207
- Fracca C, de Leoni M, Asnicar F, Turco A (2022) Estimating activity start timestamps in the presence of waiting times via process simulation. *Caise*, Springer, LNCS 13295:287–303
- Francescomarino CD, Ghidini C, Maggi FM, Milani F (2018) Predictive process monitoring methods: which one suits me best? In: BPM, Springer, Heidelberg, LNCS, vol 11080, pp 462–479
- Ganesha K, Dhanush S, SM SR (2017a) An approach to fuzzy process mining to reduce patient waiting time in a hospital. In: 2017 international conference on innovations in information, embedded and communication systems (ICIECS), IEEE, pp 1–6
- Ganesha K, Supriya KV, Soundarya M (2017b) Analyzing the waiting time of patients in hospital by applying heuristics process miner. In: 2017 international conference on inventive communication and computational technologies (ICICCT), pp 500–505
- Gerhardt R, Valiati JF, dos Santos JVC (2018) An investigation to identify factors that lead to delay in healthcare reimbursement process: a Brazilian case. *Big Data Res* 13:11–20
- Goel K, Leemans SJJ, Martin N, Wynn MT (2022) Quality-informed process mining: A case for standardised data quality annotations. *ACM Trans Knowl Discov Data* 16(5):97:1–97:47
- Graafmans T, Türetken O, Poppelaars H, Fahland D (2021) Process mining for six sigma. *Bus Inf Syst Eng* 63(3):277–300
- Grabis J (2014) Application of predictive simulation in development of adaptive workflows. In: WSC, IEEE/ACM, pp 996–1004
- Gupta M, Sureka A, Padmanabhuni S (2014) Process mining multiple repositories for software defect resolution from control and organizational perspective. In: MSR, ACM, pp 122–131
- Hompes B, Buijs JCAM, van der Aalst WMP (2016) A generic framework for context-aware process performance analysis. *OTM conferences*, LNCS 10033:300–317
- Jaisook P, Premchaiswadi W (2015) Time performance analysis of medical treatment processes by using disco. In: 2015 13th international conference on ICT and knowledge engineering (ICT & knowledge engineering 2015), pp 110–115
- Keele S et al (2007) Guidelines for performing systematic literature reviews in software engineering. Tech. rep., Technical report, ver. 2.3 ebse technical report. ebse
- Khan N, Ali Z, Ali A, McClean SI, Charles D, Taylor PN, Nauck DD (2019) A generic model for end state prediction of business processes towards target compliance. *SGAI conf*, Springer, Heidelberg, LNCS 11927:325–335
- Klijn EL, Fahland D (2019) Performance mining for batch processing using the performance spectrum. *Business process management workshops*, Springer, Heidelberg, LNBIP 362:172–185
- Kubrak K, Milani F, Nolte A (2022a) Process mining for process improvement - an evaluation of analysis practices. *RCIS*, Springer, LNBIP 446:214–230
- Kubrak K, Milani F, Nolte A, Dumas M (2022b) Prescriptive process monitoring: Quo vadis? *PeerJ Comput Sci* 8:e1097
- Kubrak K, Milani F, Nolte A (2023) A visual approach to support process analysts in working with process improvement opportunities. *Bus Process Manag J* 29(8):101–132
- Lamghari Z, Radgui M, Saidi R, Rahmani MD (2019) Predictive process monitoring related to the remaining time dimension: a value-driven framework. In: 2019 1st international conference on smart systems and data science (ICSSD), IEEE, pp 1–6
- Lashkevich K, Milani F, Chapela-Campa D, Dumas M (2022) Data-driven analysis of batch processing inefficiencies in business processes. *RCIS*, Springer, Heidelberg, LNBIP 446:231–247
- Lashkevich K, Milani F, Chapela-Campa D, Suvorau I, Dumas M (2023) Why am I waiting? Data-driven analysis of waiting times in business processes. *Caise*, Springer, Heidelberg, LNCS 13901:174–190
- Leemans M, van der Aalst WMP, van den Brand MGJ (2018) Hierarchical performance analysis for process mining. In: ICSSP, ACM, pp 96–105
- Leonardi G, Striani M, Quaglini S, Cavallini A, Montani S (2018) Leveraging semantic labels for multi-level abstraction in medical process mining and trace comparison. *J Biomed Inform* 83:10–24
- Low WZ, Weerdt JD, Wynn MT, ter Hofstede AHM, van der Aalst WMP, vanden Broucke SKLM (2014) Perturbing event logs to identify cost reduction opportunities: a genetic algorithm-based approach. In: IEEE congress on evolutionary computation, IEEE, pp 2428–2435
- Mannhardt F, Landmark AD (2019) Mining railway traffic control logs. *Transp Res Procedia* 37:227–234
- Mannhardt F, Arnesen P, Landmark AD (2019) Estimating the impact of incidents on process delay. In: ICPM, IEEE, pp 49–56
- Mans R, Reijers HA, van Genuchten M, Wismeijer D (2012) Mining processes in dentistry. In: IHI, ACM, pp 379–388
- Mansar SL, Reijers HA (2007) Best practices in business process redesign: use and impact. *Bus Process Manag J* 13(2):193–213
- Martin N, Depaire B, Caris A (2015) Using event logs to model interarrival times in business process simulation. *Business process management workshops*, Springer, Heidelberg, LNBIP 256:255–267
- Milani F, Maggi FM (2018) A comparative evaluation of log-based process performance analysis techniques. *BIS*, Springer, Heidelberg, LNBIP 320:371–383
- Milani F, Lashkevich K, Maggi FM, Di Francescomarino C (2022) Process mining: a guide for practitioners. *International*

- conference on research challenges in information science. Springer, Heidelberg, pp 265–282
- Nguyen H, Dumas M, ter Hofstede AHM, Rosa ML, Maggi FM (2016) Business process performance mining with staged process flows. Caise, Springer, Heidelberg, LNCS 9694:167–185
- Nogayama T, Takahashi H (2015) Estimation of average latent waiting and service times of activities from event logs. *BPM*, Springer, Heidelberg, LNCS 9253:172–179
- Okoli C (2015) A guide to conducting a standalone systematic literature review. *Commun Assoc Inf Syst* 37:43
- Pan Y, Zhang L (2021) Automated process discovery from event logs in BIM construction projects. *Autom Constr* 127(103):713
- Pang J, Xu H, Ren J, Yang J, Li M, Lu D, Zhao D (2021) Process mining framework with time perspective for understanding acute care: a case study of AIS in hospitals. *BMC Med Inform Decis Mak* 21(1):354
- Park G, Song M (2020) Predicting performances in business processes using deep neural networks. *Decis Support Syst* 129
- Petitdemange E, Lamine E, Fontanili F, Laurus M (2020) Enhancing emergency call centers' performance through a data-driven simulation approach. In: *ISCRAM*, *ISCRAM Digital Library*, pp 218–227
- Pika A, van der Aalst WMP, Fidge CJ, ter Hofstede AHM, Wynn MT (2013) Profiling event logs to configure risk indicators for process delays. Caise, Springer, Heidelberg, LNCS 7908:465–481
- Pla A, Gay P, Meléndez J, López B (2011) Petri net based agents for coordinating resources in a workflow management system. In: *ICAART* (1), *SciTePress*, pp 514–523
- Porouhan P, Premchaiswadi W (2018) Behavioral performance evaluation and emotion analytics of a MOOC course via fuzzy modeling. In: 2018 16th international conference on ICT and knowledge engineering (ICT & KE), IEEE, pp 1–8
- Premchaiswadi W, Porouhan P (2015) Process modeling and bottleneck mining in online peer-review systems. *SpringerPlus* 4(1):1–18
- Pufahl L, Meyer A, Weske M (2014) Batch regions: process instance synchronization based on data. In: *EDOC*, IEEE Computer Society, pp 150–159
- Rahardianto R, Sarno R, Budiawati GI (2018) Performance time evaluation of domestic container terminal using process mining and PERT. In: 2018 international seminar on application for technology of information and communication, IEEE, pp 469–475
- Rama-Maneiro E, Vidal JC, Lama M (2023) Deep learning for predictive business process monitoring: review and benchmark. *IEEE Trans Serv Comput* 16(1):739–756
- Randolph J (2007) A guide to writing the dissertation literature review. *Pract Assess Res Eval* 14
- Reijers H, Mansar S (2005) Best practices in business process redesign: an overview and qualitative evaluation of successful redesign heuristics. *Omega* 33(4):283–306. <https://doi.org/10.1016/j.omega.2004.04.012>
- Rogge-Solti A, Weske M (2015) Prediction of business process durations using non-markovian stochastic petri nets. *Inf Syst* 54:1–14
- Rojas E, Cifuentes A, Burattin A, Munoz-Gama J, Sepúlveda M, Capurro D (2018) Analysis of emergency room episodes duration through process mining. *Business process management workshops*, Springer, Heidelberg, LNBIP 342:251–263
- Salimifard K, Hosseini SY, Moradi MS (2013) Improving emergency department processes using coloured petri nets. In: *PNSE+—ModPE*, *CEUR-WS.org*, *CEUR Workshop Proceedings*, vol 989, pp 335–349
- dos Santos GA, Southier LFP, Scalabrin EE (2020) Method to reduce lead-time of business process discovered. In: *CISP-BMEI*, IEEE, pp 840–845
- Satitharoenmuang C, Porouhan P, Nammakhunt A, Saguansakiyotin N, Premchaiswadi W (2017) Benchmarking efficiency of children's garment production process using alpha and ILP replay techniques. In: 2017 15th international conference on ICT and knowledge engineering (ICT & KE), IEEE, pp 1–7
- Senderovich A, Rogge-Solti A, Gal A, Mendling J, Mandelbaum A, Kadish S, Bunnell CA (2015) Data-driven performance analysis of scheduled processes. *BPM*, Springer, Heidelberg, LNCS 9253:35–52
- Senderovich A, Shleyfman A, Weidlich M, Gal A, Mandelbaum A (2016) P<sup>3</sup> -folder: optimal model simplification for improving accuracy in process performance prediction. *BPM*, Springer, Heidelberg, LNCS 9850:418–436
- Senderovich A, Weidlich M, Gal A (2019) Context-aware temporal network representation of event logs: model and methods for process performance analysis. *Inf Syst* 84:240–254
- Singh S, Verma R, Koul S (2021) A collaborative method for simultaneous operations: case of an eye clinic. *Opsearch* 59(2):711–731
- Taymouri F, Rosa ML, Dumas M, Maggi FM (2021) Business process variant analysis: survey and classification. *Knowl Based Syst* 211(106):557
- Teinemaa I, Dumas M, Rosa ML, Maggi FM (2019) Outcome-oriented predictive process monitoring: review and benchmark. *ACM Trans Knowl Discov Data* 13(2):17:1–17:57
- Thomas L, V MKM, Basava A, Puttanna VK (2015) An optimal process model for a real time process. In: *Ataед@petri nets/acsd*, *CEUR-WS.org*, *CEUR Workshop Proceedings*, vol 1371, pp 117–131
- Toosinezhad Z, Fahland D, Köroglu Ö, van der Aalst WMP (2020) Detecting system-level behavior leading to dynamic bottlenecks. In: *ICPM*, IEEE, pp 17–24
- Tridalestari F, Mustafid M, Warsito B, Wibowo A, Prasetyo H (2022) Analysis of e-commerce process in the downstream section of supply chain management based on process and data mining. *Ing Syst Inf* 27(1):81–91
- van der Aalst WM (2011) Process discovery: an introduction. *Process mining*. Springer, Heidelberg, pp 125–156
- van der Aalst WMP (2016) Process mining - data science in action, vol, 2nd edn. Springer, Heidelberg
- van der Aalst WMP, Low WZ, Wynn MT, ter Hofstede AHM (2015) Change your history: learning from event logs to improve processes. In: *CSCWD*, IEEE, pp 7–12
- van der Aalst WMP, Rosa ML, Santoro FM (2016) Business process management - don't forget to improve the process! *Bus Inf Syst Eng* 58(1):1–6
- Verenich I, Dumas M, Rosa ML, Maggi FM, Teinemaa I (2019) Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring. *ACM Trans Intell Syst Technol* 10(4):34:1–34:34
- Yampaka T, Chongstitvatana P (2016) An application of process mining for queueing system in health service. In: 2016 13th international joint conference on computer science and software engineering (JCSSE), IEEE, pp 1–6
- Zeng Q, Liu C, Duan H, Zhou M (2020) Resource conflict checking and resolution controller design for cross-organization emergency response processes. *IEEE Trans Syst Man Cybern Syst* 50(10):3685–3700